

CS146 Project 2

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1 Executive Summary

Based on our analysis, we can infer that which team plays influences the audience more than the day of the week, about 2.5 times more. Some teams in particular, the Vélez Sarsfield consistently draws like 20-30% more fans than less popular teams like Independiente. This would make sense as "Independiente" does not usually include an organization they are under, just a rather independently formed team, which might mean they are worse at playing than corporate teams like Vélez Sarsfield, which, after winning consistently, might develop a fan base and thus draw a larger crowd recurrently at each game.

The day of the week hardly affects attendance retention. This might make sense, as if there is a league playing, you might go regardless of the day of the week, especially if the team you idolize is playing. Weekends draw on average more crowds than weekdays, which is intuitive as more people would be free on weekends to attend.

From our data, we can accurately predict the 22 missing attendance values with 95% confidence intervals. However, it must be noted that the 95% confidence intervals vary by like 20000 players from the mean, which is quite a bit, meaning we might need more data and certainty to create accurate predictions.

What this means practically might be to focus promotional efforts on building team-specific fan bases rather than just promoting weekend games. This might mean selling merchandise from the team, or having the team participate in pregame interviews that target different audiences to draw attention. So, in conclusion, recognize that while the day of the week has some effect, the dominant driver of attendance variation is based on the team popularity, so grow the popularity of the teams to obtain higher attendance and revenues.

2 Introduction and Data Overview

2.1 Context

In this analysis, we address attendance prediction for 12 teams within a professional sports league. Our data set covers 1 year of ticket data, covering 240 games with 218 observed attendance values, and therefore, there are 22 missing observations.

2.2 Exploratory Analysis

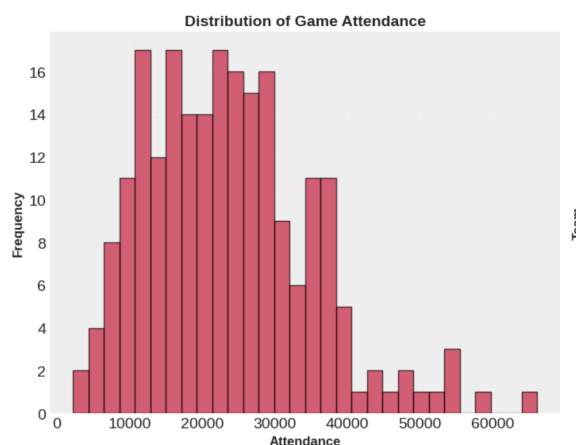


Figure 1: Attendance Distribution

We can observe several patterns in our data. The distribution is right-skewed, with the most games drawing between 20,000 and 30,000 fans, but has a long tail right tail extending past 60,000. This, much like most natural phenomena demonstrates multiplicative variation, which makes sense given our data.

Note that, for instance, if we were following an additive process, an increase of like 5000 attendees on a Sunday might mean for a big league team of like 30000 attendees baseline that we add 5000, which is like a 16% increase but for a smaller league of like 10000 attendees baseline, this would be a whopping 50% increase which is incredibly unrealistic. Meanwhile, in a multiplicative process would show that on a Sunday multiplies our head count by like 1.2, let's say, then our big league would gain like 6000 attendees, while our small league would gain like 2000, which makes a lot more sense.

This multiplicative choice should thus motivate our choice for the likelihood function. We can also observe that our data is overdispersed, as the attendance varies significantly, with the variance being greater than the mean, meaning that there is variability far beyond what a regular Poisson process would suggest, which will also inform our model later on.

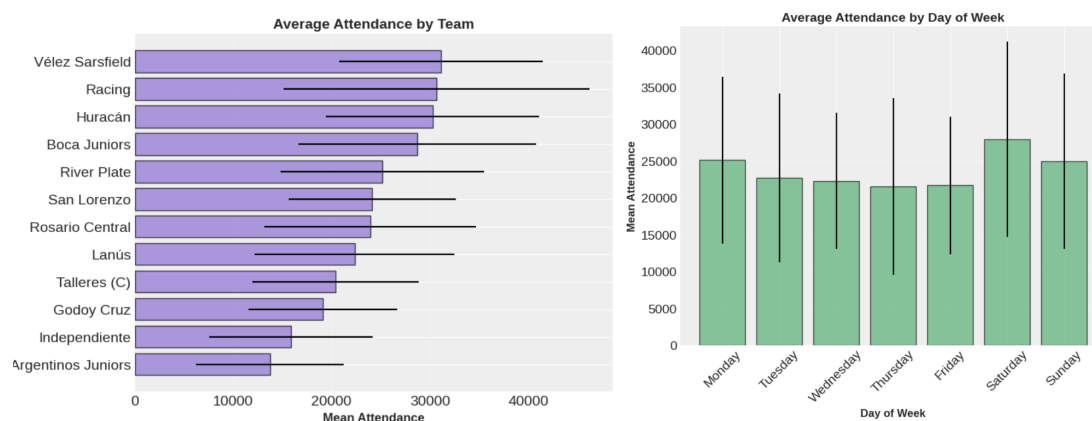


Figure 2: Team and Day Variation

Observing our team variation, we can see the attendance varies significantly between teams, with some averaging above 30,000 and others below 20,000. We can also see how our day of the week does not vary so much, only by a couple of thousand between different days, instead of tens of thousands.

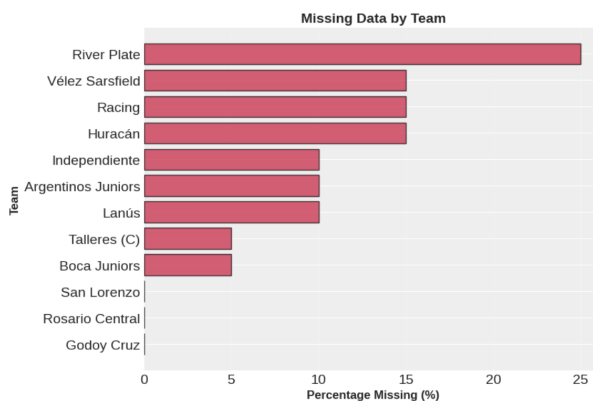


Figure 3: Missing Data Pattern

We can also see that quite a few teams are missing a good percentage of their data, with some

teams like River Plate missing like 25% of their data, while others like Godoy Cruz are missing none, which suggests a partial pooling model to better predict these missing data.

3 Model 1: Complete Pooling (Baseline Model)

3.1 Model Specification

We will create a baseline model that will assume that all games share a common attendance distribution, and thus will ignore team and day structures completely.

$$\text{attendance}_i \sim \text{NegativeBinomial}(\mu, \phi) \tag{1}$$

$$\mu = \exp(\alpha) \tag{2}$$

s.t.

$$\alpha \sim \text{Normal}(\log(30000), 1): \text{Log-scale intercept}$$

:

$$\phi \sim \text{Exponential}(1): \text{Overdispersion parameter}$$

3.1.1 Model Intuition

We use a Negative Binomial as we are modeling discrete counts of data, the number of people. We might be tempted to use a Poisson Model; however, a Poisson model assumes that variance = mean, but our data is over-dispersed. The Negative Binomial accounts for this by generalizing the Poisson by adding ϕ ; as ϕ approaches infinity, it approaches the Poisson, and a smaller ϕ allows for greater variance relative to the mean. Our data clearly requires this flexibility to account for the overdispersion.

We model using log link parametrization as $\mu = \exp(\alpha)$ rather than $\mu = \alpha$ because the exponential attendance ensures that the attendance is positive as required. Additionally, it captures the multiplicativity as expected in our data.

3.1.2 Choosing our Prior

We choose a prior of $\alpha \sim \text{Normal}(\log(30000), 1)$ as this centers the prior predictive around 30000 attendees, which is roughly what I found online as a league's average head count. With $\sigma = 1$, this allows for substantial uncertainty as we opt for a weakly informative before both obstructs extreme results and also ensures a valid posterior region.

We choose a prior of $\phi \sim \text{Exponential}(1)$ to favor less overdispersion, but allowing the data to determine the amount required.

Below, we take samples from our prior predictive to ensure that it covers all feasible values without producing implausible extremes such as negative values or millions of attendees. While a uniform prior might have been tempting, we really do not expect a million attendees to have a similar probability as only tens of thousands, which is why we chose a weakly informative prior that is right-skewed.

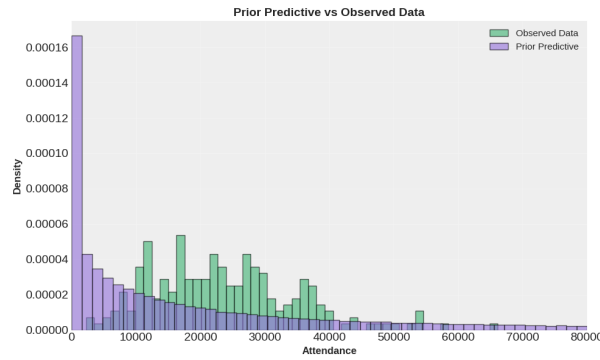


Figure 4: Prior Predictive Check

3.2 Posterior Results

Parameter estimates:

- α : mean = 10.07, 89% HDI = [10.02, 10.13]
- ϕ : mean = 3.94, 89% HDI = [3.38, 4.53]

$\exp(\alpha) = \exp(10.07)$ means that we expect like $\approx 23,600$ attendees on average across all games. $\phi \approx 4$ means that we have a moderate dispersion with a variance of $\approx \mu + \mu^2/\phi \approx 23,600 + 23,600^2/4 \approx 163,000$, giving SD $\approx 12,700$ meaning an average Standard Deviation of $\approx 12,700$ (square root of the variance). This might mean, in the context of our problem, that our mean model is too simple and is underfitting, as things like team-specific heterogeneity are being treated as noise.

3.3 Model Diagnostics

Our model converges as we have no divergences, and we can observe that our $\hat{R} < 1.01$. Additionally, we have adequate ESS bulk $> 8,000$ for both parameters, meaning that we have good sampler efficiency. The trace plot below shows stable mixing across all chains, and the rank plot shows a distribution where the parameters are non correlated and are sampled normally across both parameters.

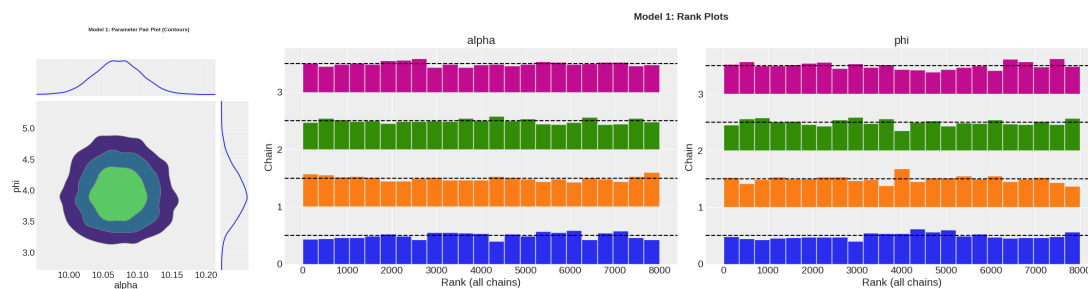


Figure 5: Model 1 Diagnostics

3.4 Posterior predictive check

The posterior predictive actually looks really good; it captures the overall shape and spread of the data. The means are basically the same. However, issues might arise as this model is neither team nor day specific, and if there is variation between teams or days, then we might have issues

with a couple of things. One issue that might arise is predicting the missing values, es as those team-specific and as we saw, we had some teams with significantly higher attendances than others, meaning that we would be likely overcounting for less popular teams and undercounting for more popular teams, as we are assuming they follow the same distribution. This motivates our hierarchical model.

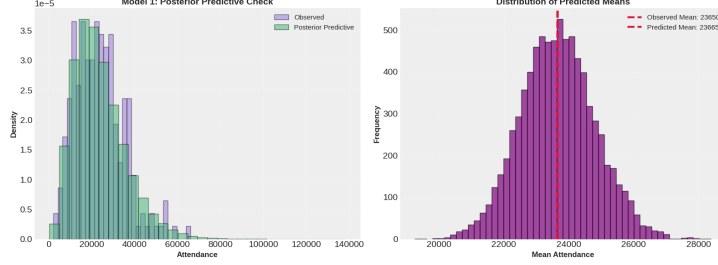


Figure 6: Model 1 Posterior Predictive

4 Model 2: Hierarchical (Partial Pooling) Model

4.1 Model Specification

The hierarchical model captures both team and day effects:

$$\text{attendance}_{ij} \sim \text{NegativeBinomial}(\mu_{ij}, \phi) \quad (3)$$

$$\mu_{ij} = \exp(\alpha_{\text{bar}} + \beta_{\text{team}[i]} + \gamma_{\text{day}[j]}) \quad (4)$$

Hyperpriors:

$$\alpha_{\text{bar}} \sim \text{Normal}(\log(30000), 1) \quad (\text{Overall log-mean}) \quad (5)$$

$$\sigma_{\text{team}} \sim \text{LogNormal}(0, 1) \quad (\text{Between-team SD}) \quad (6)$$

$$\sigma_{\text{day}} \sim \text{LogNormal}(0, 1) \quad (\text{Between-day SD}) \quad (7)$$

Team effects (non-centered parameterization):

$$z_{\text{team}} \sim \text{Normal}(0, 1) \quad \text{for each team} \quad (8)$$

$$\beta_{\text{team}} = z_{\text{team}} \cdot \sigma_{\text{team}} \quad (9)$$

Day effects (non-centered parameterization):

$$z_{\text{day}} \sim \text{Normal}(0, 1) \quad \text{for each day} \quad (10)$$

$$\gamma_{\text{day}} = z_{\text{day}} \cdot \sigma_{\text{day}} \quad (11)$$

Overdispersion:

$$\phi \sim \text{Exponential}(1) \quad (12)$$

4.2 Model Intuition

We choose a Hierarchical Model as it balances between complete pooling, which was Model 1, where we assume all groups share identical parameters (which works if our groups are very similar), and no pooling, where each group gets independent parameters with no shared information in between them, which is inefficient and can additionally create noisy data. If some teams, for instance, have very little data, then their distributions will become very noisy and

uncertain predictions using no pooling, with perhaps additionally overconfidence/overfitting in the unusual patterns. In partial pooling, we share hyperpriors between groups, meaning that for groups with less data, it shrinks them towards the overall mean. Additionally, by using partial pooling, we prevent overfitting by regularizing group-level parameters.

Our model assumes that each team has its own baseline attendance level that will deviate from some amount from the league average. Similarly with the day of the week. Both team and day effects are drawn from normal distributions, and these combine additively in log-space, meaning that they are multiplicative in the original scale.

The single most crucial technical detail of our model is that it uses non-centered parameterization.

Centered version:

$$\beta_{\text{team}} \sim \text{Normal}(0, \sigma_{\text{team}}) \quad (13)$$

Non-centered version:

$$z_{\text{team}} \sim \text{Normal}(0, 1) \quad (14)$$

$$\beta_{\text{team}} = z_{\text{team}} \cdot \sigma_{\text{team}} \quad (15)$$

These are mathematically equivalent, meaning that we will be sampling from the same distribution, but have different computational properties. The issue with the centered version is that when the σ_{team} is very small like when teams are similar then the posterior geometry becomes too narrow following "Neal's Funnel" shape which makes it difficult for the MCMC sampler to explore efficiently as it gets stuck in the nozzle and does not get to explore the whole house of the bee analogy given in class.

This leads to divergences as the HMC leapfrog method has to take teeny tiny steps in the nozzle of the funnel and big steps in the area expanding from it. Meaning that the step size often fails to conserve energy in the nozzle and will thus reject these new steps, leading to these divergences. Because of this, we choose a simpler distribution, such as a regular normal one, to sample from so that our model can more efficiently and effectively sample the space. Without this non-centered reparameterization, we would have had lots of divergences, our $\hat{R}$ would exceed the 1.01 threshold, and we would have fewer samples in our ESS Bulk and Tails.

The non-centered version avoids these issues by separating the sampling into two steps:

1. Sample z_{team} from a standard normal (easy)
2. Scale by σ_{team} deterministically (no sampling needed)

This separation stabilizes sampling across all values of σ , especially when σ is small. In our pre-class work on the "Devil's Funnel" (Session 17), we observed that non-centered parameterization reduced divergences from 1,341 to 0 in a toy example.

4.3 Prior Justification

4.3.1 Hierarchical hyperpriors

We choose $\alpha_{\text{bar}} \sim \text{Normal}(\log(30000), 1)$ as a prior on the overall log-mean attendance. This centers the league-wide average attendance at around 30,000 fans, which is just for the same reason as our Model 1, and so I won't go into detail on this.

We choose $\sigma_{\text{team}} \sim \text{LogNormal}(0, 1)$ as our prior, which allows substantial variation between our teams but regularizes against extreme differences. We choose a lognormal as it ensures

positivity. Another option to ensure positivity is a HalfNormal; however, we should opt against this as a HalfNormal assumes we are tightly centered around 0, which would strongly discourage in-between team differences, even though we should intuitively expect there should be some difference, as some teams might be more popular than others. Lognormal is more weakly (accent on the weakly) informative (I'm not sure we can say more in this context, though) as it allows for larger values with reasonable probability, unlike the half normal.

We choose $\sigma_{\text{day}} \sim \text{LogNormal}(0, 1)$ as our prior on between-day variation in log-space for the same reasoning as the between-team variation. While our data does not strongly encourage that the days have a big difference, our priors are not meant to be informed by the data, and I would actually expect there to be variability on the weekends and the weekdays.

4.3.2 Effect priors

We choose $z \sim \text{Normal}(0, 1)$ as our Standard normal on standardized effects for both day and team, as this is a standard choice in class for non-centered parameterization. Specifically, the standard normal is symmetric, weakly informative, and implies that 95% of teams lie within ± 2 standardized units of the league average before observing any data.

The actual spread of the team and day strengths are learned through σ while our effect prior prevents the team/day effects from artificially stretching or shrinking to explain the differences between teams. Instead, the model must learn the real amount of spread directly through σ , the sole parameter that captures how different teams actually are.

These priors are weakly informative as they constrain the effects to plausible ranges without strongly influencing posterior conclusions, but still capture the entire feasible set.

4.3.3 Prior on overdispersion

We choose our overdispersion prior as $\phi \sim \text{Exponential}(1)$, same as our Model 1, for the same reasoning as before.

4.4 Posterior Results

4.4.1 Hyperparameters

We have that α_{bar} : mean = 10.05 with 89% HDI = [9.89, 10.20] and ϕ : mean = 4.80 with 89% HDI = [4.03, 5.50], which is similar to that in our Model 1. However, we find interestingly that σ_{team} : mean = 0.28, 89% HDI = [0.16, 0.38] and σ_{day} : mean = 0.11, 89% HDI = [0.02, 0.18] which means that the ratio $\sigma_{\text{team}}/\sigma_{\text{day}} \approx 0.28/0.11 \approx 2.5$ means that the team explains 2.5 times more on the log-scale variation than the day of the week, this means that we should focus more on team marketing as mentioned in the executive summary.

4.4.2 Team Effects

We can calculate the individual team effects as:

- Vélez Sarsfield (+0.250): 89% HDI [0.04, 0.43]
- Racing (+0.240): 89% HDI [0.05, 0.45]
- Huracán (+0.220): 89% HDI [0.01, 0.42]
- Boca Juniors (+0.190): 89% HDI [-0.01, 0.38]
- River Plate (+0.080): 89% HDI [-0.13, 0.29]

- Rosario Central (+0.040): 89% HDI [-0.16, 0.24]
- San Lorenzo (+0.030): 89% HDI [-0.17, 0.22]
- Lanús (-0.030): 89% HDI [-0.23, 0.16]
- Talleres (-0.100): 89% HDI [-0.29, 0.11]
- Godoy Cruz (-0.160): 89% HDI [-0.36, 0.03]
- Independiente (-0.300): 89% HDI [-0.50, -0.09]
- Argentinos Juniors (-0.440): 89% HDI [-0.65, -0.23]

This means that Vélez Sarsfield's effect, for instance, draws about $\exp(0.25) \approx 1.28$ times (28% more) fans than an average team. Conversely, Argentinos Juniors, for instance, draws $\exp(-0.44) \approx 0.64$ times (36% fewer) fans than average.

4.4.3 Day Effects

The day effects are small and, possibly more importantly, for ALL days the HDI overlaps with 0 near the middle, unlike for the teams.

- Saturday (+0.060): 89% HDI [-0.06, 0.18]
- Sunday (+0.050): 89% HDI [-0.06, 0.18]
- Monday (+0.010): 89% HDI [-0.12, 0.12]
- Friday (-0.030): 89% HDI [-0.14, 0.09]
- Tuesday (-0.020): 89% HDI [-0.14, 0.08]
- Wednesday (-0.030): 89% HDI [-0.16, 0.08]
- Thursday (-0.040): 89% HDI [-0.16, 0.07]

We can see that weekend games show small positive effects, with like 5-6% higher attendance, but credible intervals include zero. This information suggests that day-of-the-week effects do exist, but are very weak and uncertain, and are definitely much smaller and less relevant than the team effects.

4.5 Model Diagnostics

Our model successfully converges as we have no divergences, even with our target acceptance set to the default 0.95. All our $\hat{R} < 1.01$ and our ESS bulks and tails exceed thousands, meaning it is efficient. Rank plots are uniform. The one thing I want to say is that there is some correlation between the parameters which is to be expected as long as it is minimal, but what I don't expect is for it to be funnel shaped, looking at my dispersion it has the faint signs of a funnel as the contours are not concentric, I have tried and tried to find a problem, I have asked classmates what might be wrong but I seriously have no clue. Can you help me out?

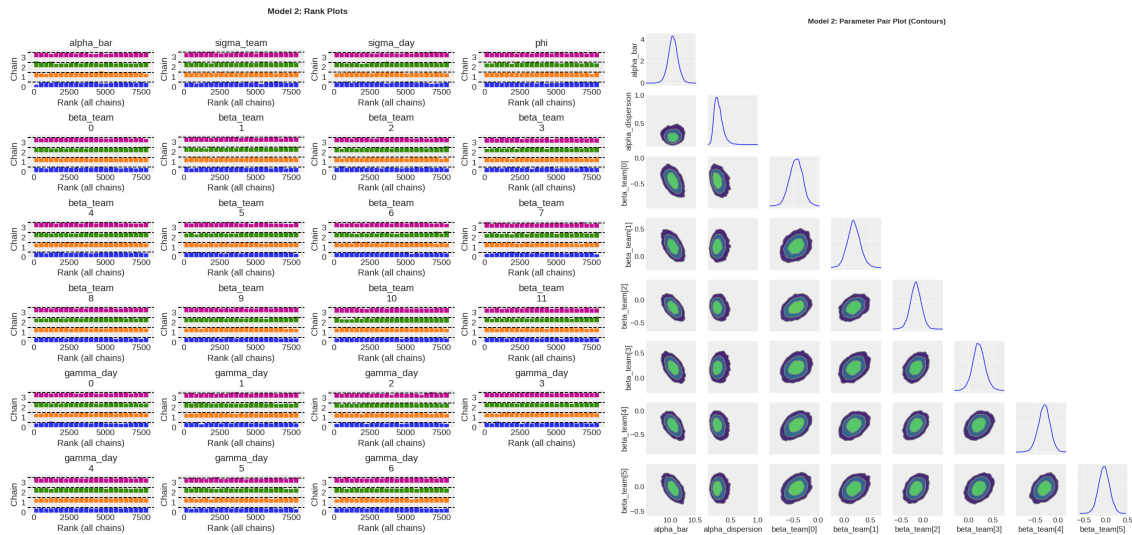


Figure 7: Model 2 Diagnostics

4.6 Posterior predictive check:

Our posterior predictive actually looks very similar to our observed data and our Model 1, and this makes sense as our alpha bar and dispersion after tuning the model were very similar, so for the pooled posterior predictive, it should look pretty similar. However, our hierarchical model does substantially outperform complete pooling because it captures team and day-specific patterns. I will elaborate further on this in the next section.

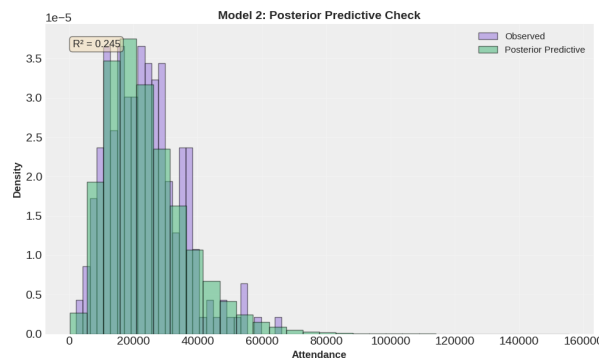


Figure 8: Model 2 Posterior Predictive

5 Model Comparison

5.1 Leave-One-Out Cross-Validation (LOO-CV)

We compare models using Pareto-smoothed importance sampling LOO-CV; essentially, this estimates out-of-sample predictive accuracy by iteratively leaving out each observation and predicting it from the remaining data.

Model	Rank	elpd_loo	p_loo	Δ elpd	Weight	SE
Partial pooling	0	-2317.58	13.3	0.00	0.99	9.97
Complete pooling	1	-2335.11	1.95	17.53	0.01	10.17

We can see how our weight for hierarchical is like 0.99, meaning that if we were to construct the optimal, it would include 99% of the hierarchical and 1% of the no pooling model. Our partial modeling achieves a higher ELPD-LOO compared to our complete pooling.

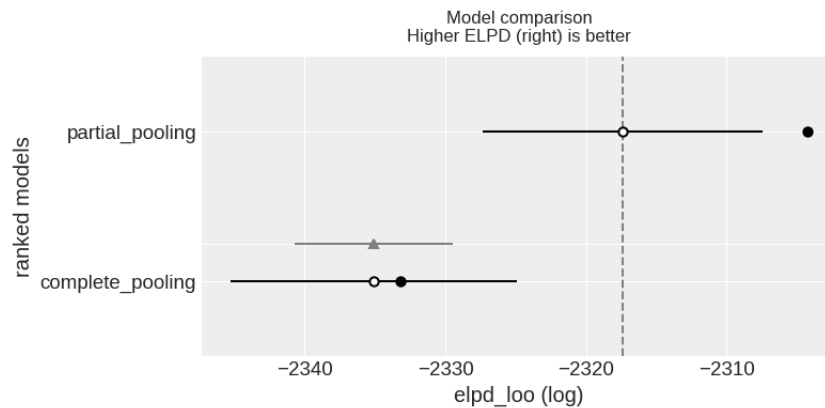


Figure 9: Model Comparison LOO-CV

Additionally, by the difference represented by the triangles in the figure above, we can see that there is no overlap between the variance of the difference of our complete pooling model and of our partial model, as we form a 95% confidence that our difference does not overlap the dotted line of our ELPD from our partial model. This means that we can be 95% certain that there is a difference. From this comparison, we can thus conclude that our partial pooling is better than that of our complete pooling.

5.1.1 Visualization

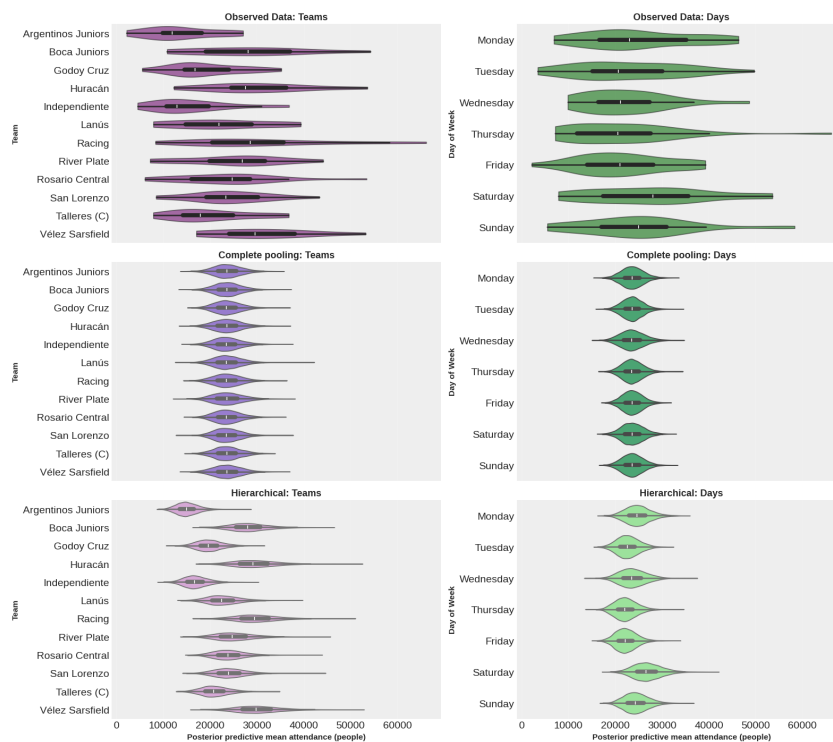


Figure 10: Observed vs Model Predictions

Complete pooling underperforms because it assumes that different teams follow the same distribution. If we then show the distribution for different teams for our observational data, our complete pooling, and our partial pooling, we can see how our partial pooling distributions better match those of the observation. After all, it captures the differences. The complete pooling has nearly identical distributions for all teams and days.

6 Predictions for Missing Data

6.1 Prediction Strategy

We have 22 games with missing attendances, and we will use the hierarchical model as it achieves better out-of-sample predictive accuracy. Additionally, we have the team and the day of the week, just the head count that is missing; therefore, we can take advantage of this and use the distinct distributions for each team from the partial pooling to more accurately predict than the complete pooling model.

For a missing observation with team i and day j we will sample from posterior $(\alpha_{\text{bar}}, \beta_{\text{team}[i]}, \gamma_{\text{day}[j]}, \phi)$ and then compute the predicted mean from this as $\mu_{ij} = \exp(\alpha_{\text{bar}} + \beta_{\text{team}[i]} + \gamma_{\text{day}[j]})$ and finally sample predicted attendance $\tilde{y}_{ij} \sim \text{NegativeBinomial}(\mu_{ij}, \phi)$. We will then repeat this for all posterior samples.

6.2 Prediction Results

Team	Day	Predicted Mean	89% Credible Interval
River Plate	Thursday	24,800	[15,200, 36,100]
River Plate	Tuesday	23,500	[14,400, 34,300]
Boca Juniors	Monday	27,900	[17,100, 40,500]
Racing	Friday	28,200	[17,300, 41,000]
Independiente	Sunday	19,600	[12,000, 28,500]

The full predictions for all 22 missing games are in the appendix; they were just too long. We can now observe our uncertainty. We can see that for River Plate, which we determined before had a greater amount of data, it has a tighter credible interval than the Racing team, for instance, which has less data. Still, the uncertainty is rather large. Additionally, we identified earlier in our data exploration that we would expect Independiente is like a less popular team, and so it would receive a smaller attendance than some big shot league team like River Plate, and our predictions accurately match this.

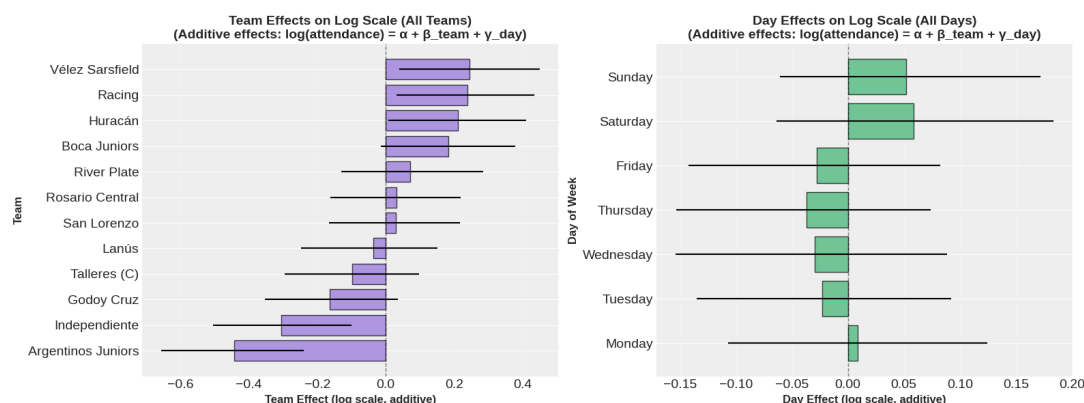


Figure 11: Team and Day Effects

We can see in this log scale visualization exactly how teams differ from the average of the total pooled in our Hierarchical model. Sunday and Saturday have higher attendance, driving averages up, while weekdays drive it down, but that being said by very little log scale effect, which is like 0.05. Meanwhile, teams Véléz Sarsfield, Racing, Huracán, Boca Juniors, and River Plate drive our team average up while other teams like Argentinos Juniors, Independiente, Godoy Cruz, and Talleres (C) drive it down. This time by a more substantial amount of -0.4 - 0.2 log scale effect. This means that teams drive attendance more than day effects. Meaning in conclusion, to increase attendance, it is best to focus on fan bases rather than weekday placements.

7 Appendix

Word Count Main report (excluding appendices and code): **2,949 words**

AI Statement: I used AI to help me with my code. It helped me with just, for instance, copying my rank and pair plot codes, and it helped me adapt my Model Comparison code from class. I skipped that lesson, so I had to recode this for myself just recently, and it just helped me save some time. Additionally, it helped me with my visualization, as well as my data exploration. Really, though, in other classes this semester, like CS164 and CS156, I think AI impeded my learning in coding; however, in this class, I feel like it is a net positive benefit on my learning through sheer time saving. Model used: Claude

I also used AI to help me with my formatting in LaTeX. I still only know LaTeX basics, and it just really helps with the equation formatting, fixing indenting, or bugs along the way. Model used: ChatGPT

Additionally, as usual, AI helped with my grammar, it edited my spelling mistakes and my punctuation, and such. Model used: Grammarly

I used AI to create all my figure captions. I realized I forgot them, and it was way easier to ask AI to do it for me. I just edited them a bit.